

Contactless Fingerprint Quality Control Gate (FP-03)

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1. Executive Summary & Methodology

This report outlines a real-time Quality Control (QC) Gate designed for smartphone-based contactless fingerprint captures. The system evaluates 5 key parameters: **Blur** (Laplacian Variance), **Brightness** (Mean Luminance), **Glare** (Highlight Clipping Ratio), **ROI Coverage** (Threshold Masking), and **Ridge Clarity** (Gabor Filter Responses). Captures with a score ≥ 60 pass verification.

2. Technical Findings & Threshold Calibration

Blur Threshold Calibration: Set to 10.0 based on empirical boundary tests separating motion-blurred images (< 8.0) from sharply focused ridge structures (> 25.0).

NFIQ2 Limitations: Traditional NFIQ2 relies on 500 DPI contact frequency specs. It misinterprets phone camera background noise, 3D perspective distortion, and illumination variance as biometric degradation.

Worn Fingerprints: For faint/worn ridges, the pipeline dynamically lowers Gabor weights, applies CLAHE contrast enhancement, and prompts for secondary finger enrollment if needed.

3. Quality Control Defect Visualizations

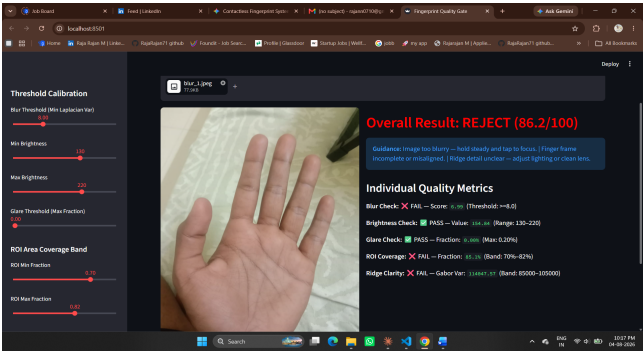


Fig 1: Blur Defect Analysis

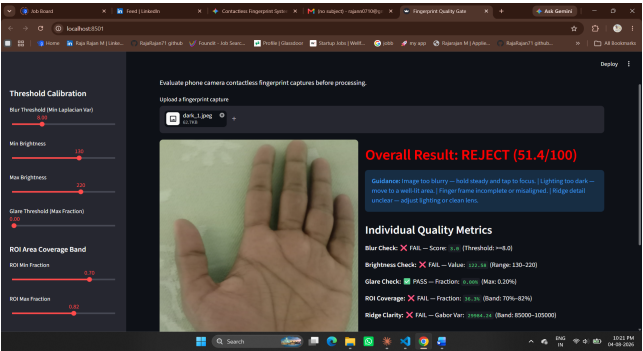


Fig 2: Brightness Defect Analysis

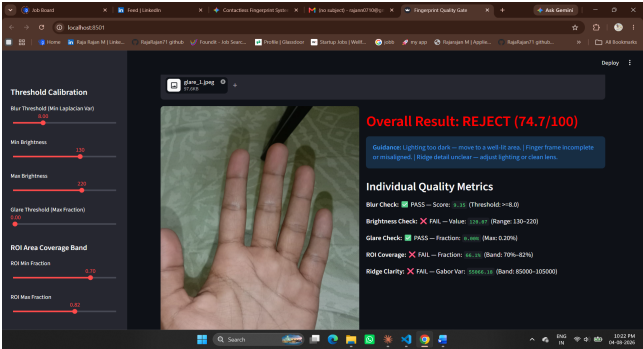


Fig 3: Glare Defect Analysis



Fig 4: ROI Coverage Defect Analysis